

Verified Family Consent for YouTube India

Making under-18 data protection DPDP-compliant —
without ever locking a teen out of learning.

Product: **YouTube India** Focus: **Children's data & parental consent**

Problem Statement

Which product, which slice of data/consent, and why it matters

The gap

YouTube India verifies age by a self-declared birthday. A 15-year-old taps through in seconds — and is then treated like an adult: behavioural profiling, activity tracking and targeted ads all run by default.

Why it matters

- **Legal risk** — DPDP Act 2023 mandates verifiable parental consent for minors and bars targeted ads / behavioural monitoring of children.
- **Child safety & trust** — Minors' data is profiled without a guardian ever being asked.
- **Scale** — A large share of India's ~460M YouTube users are students; the education use-case is core.

Assumptions made

- 1 ~15–20% of Indian YouTube accounts are under 18 (self-declared + estimated).
- 2 DigiLocker is a viable, widely-held rail to verify a guardian is 18+.
- 3 “Verifiable parental consent” is satisfied by a verified adult confirming — not by collecting ID numbers.
- 4 We design the happy path; edge cases (no parent, disputes) are noted, not built.

User & Context

Whose data is at stake — and who must act to protect it

M

PRIMARY · consent-giver

Meera, 42 — the Parent

Working mother in Pune. Wants her son to learn on YouTube but is anxious about tracking, ads and 'what the algorithm feeds him'. Low patience for complex setup.

Privacy & consent pain points

- No real way to consent or control today
- All-or-nothing: block the app or accept everything
- Withdrawing/ managing data is buried & confusing

A

SECONDARY · the child user

Aryan, 15 — the Teen

Class 10 student who uses YouTube daily for revision. His fear is being locked out mid-study while a parent 'sorts something out'.

Privacy & consent pain points

- Consent friction can cut off learning access
- Doesn't understand what data is collected
- Wants continuity, not a hard gate

Why this group: legal exposure, child-safety risk and trust all concentrate on the under-18 cohort — and the guardian, not the child, is the one who can lawfully consent.

Current Experience & DPDP Gaps

How data is handled today vs. what the Act expects

Today: self-declared birthday → full adult account → data used for recommendations, ad-profiling & tracking, under one blanket Terms-of-Service tap.

DPDP EXPECTATION	WHERE YOUTUBE FALLS SHORT TODAY	STATUS
Verifiable parental consent	Only a self-declared age; no guardian is ever verified.	GAP
Purpose limitation	Data collected to run the app is reused for ad targeting.	GAP
No profiling / targeted ads to children	Behavioural profiling & targeted ads run by default on minors.	VIOLATED
Easy withdrawal of consent	Buried deep in Google-wide account settings.	WEAK
Data minimisation	More collected than the learning experience needs.	WEAK
Rights: access • correct • erase	Exist, but scattered and not child/guardian-framed.	WEAK

Proposed Solution

Verified Family Consent — a privacy-by-default consent flow

Never lock the teen out

“Soft-pending”: the teen keeps watching a curated, safe feed while a parent verifies.

Verifiable consent, minimal data

DigiLocker confirms only that the guardian is 18+ — never the ID number.

Private by default

Tracking, profiling & targeted ads are OFF unless explicitly allowed.

No dark patterns

“Allow” and “Not now” carry equal visual weight.

Symmetric & auditable

One-tap withdrawal + a full rights centre and consent record.

The 9-step happy path (built as the clickable prototype)



Why this over alternatives: a hard block kills learning engagement · a credit-card check excludes many families and collects more data · an email-only 'I'm the parent' claim isn't verifiable. DigiLocker is government-grade, privacy-preserving and high-trust.

Metrics & Success Criteria

One North Star, supporting signals, guardrails — and what 'worked' means

NORTH STAR

Verified Consent Coverage

% of active under-18 users operating under valid, current, verified parental consent.

Target: $\geq 70\%$ within 90 days

Supporting metrics

- **Consent completion rate**
invite $\rightarrow$ verified
- **Median time-to-verify**
parent effort
- **Teen retention in pending**
no-lockout proof
- **Withdrawal rate**
trust / regret signal

Guardrails

- **Teen watch-time / DAU**
must not fall — no lockout harm
- **Parent grievance rate**
friction & confusion signal
- **Support ticket volume**
operational cost of the flow

Success criteria (measurable, 90 days post-launch)

- $\geq 70\%$ of detected under-18 accounts reach verified consent in 30 days
- Teen 7-day retention within 2% of pre-launch baseline
- Median parent verification under 3 minutes
- Zero targeted ads served to verified-minor accounts

Diagnostic Thinking

North Star drops 15% three weeks in — no release shipped. Why?

**Verified Consent
Coverage ▼ 15%**

Hypothesis tree

Instrumentation (check first)

- Consent-event logging bug
- Metrics pipeline delay / double-count
- Denominator definition changed

Denominator grew

- Exam-season influx of new under-18s
- Age-detection model now flags more minors
- Coverage % falls though numerator is flat

Funnel / completion fell

- DigiLocker outage or latency spike
- Invite email deliverability dropped
- Drop-off at a specific step

Consent lost

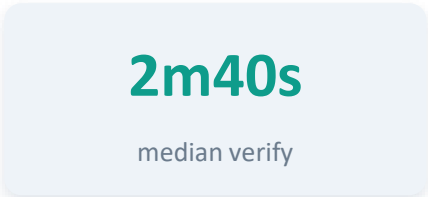
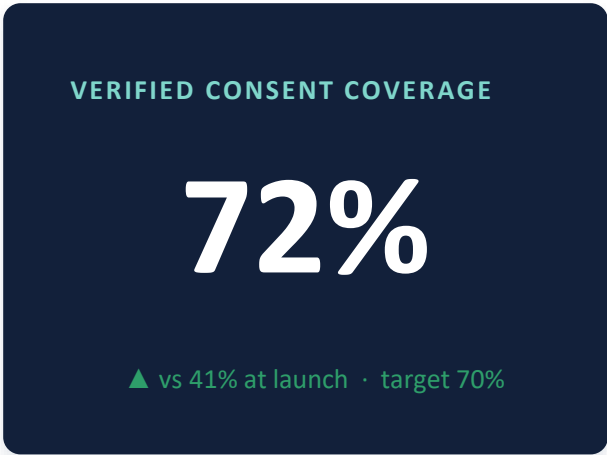
- 12-month re-confirmation wave expiring
- Withdrawal spike after a privacy news scare
- Bug invalidating stored consents

Root-cause approach

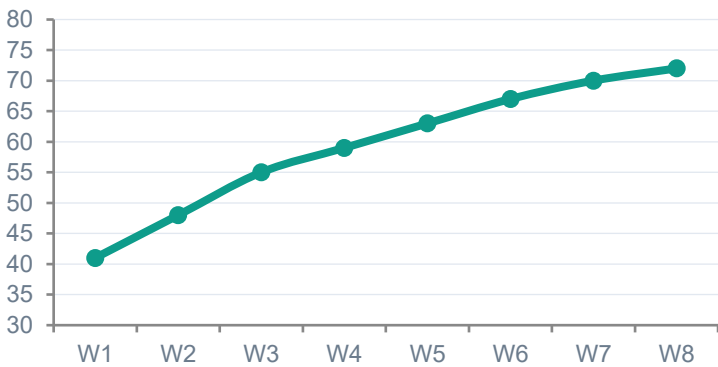
1 Confirm data integrity → 2 Segment: all users or one cohort/geo/device? → 3 Isolate the failing funnel step → 4 Check DigiLocker dependency & timeline correlation → 5 Fix, then back-fill affected consents.

Health Dashboard

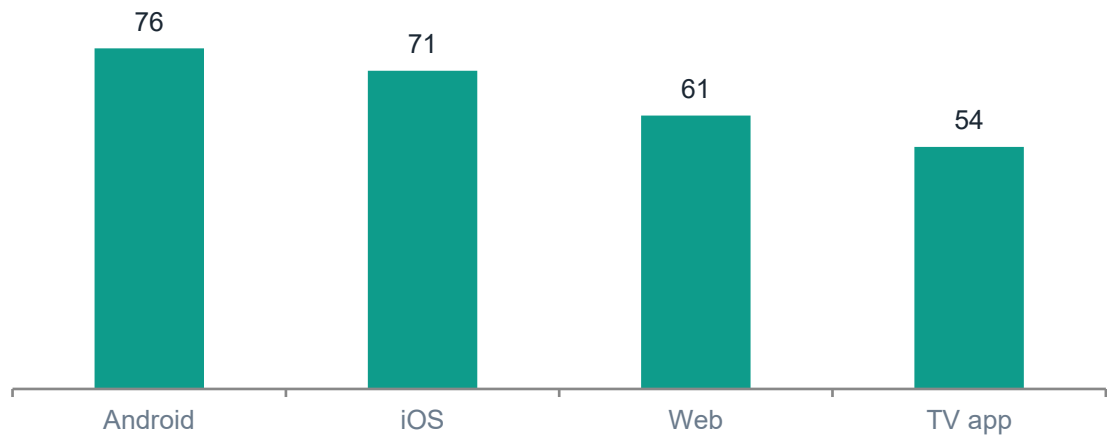
Mock data — monitoring the health of Verified Family Consent



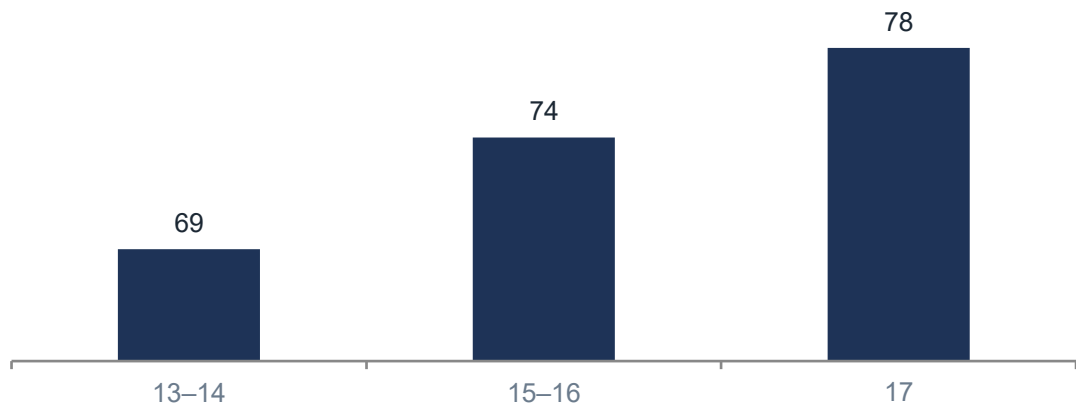
Coverage trend — 8 weeks



Segmentation — coverage by platform

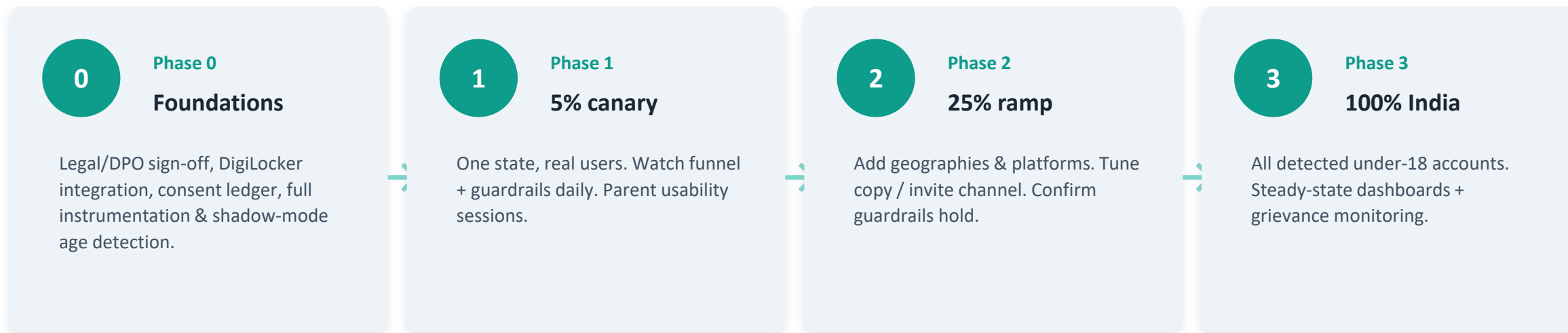


Segmentation — by age band



Rollout Plan

Ship a compliance-critical flow safely — where A/B testing does and doesn't apply



Experimentation — and its limits

Compliance is not A/B-testable. We can't lawfully withhold the consent mechanism from a control group of minors. So the **core flow ships via phased ramp + geo-staggered rollout with strict pre/post monitoring** — the phased ramp is the safety net a control group would normally provide.

We only A/B-test non-compliance-critical UX: invite copy, link vs. QR, reminder cadence, and consent-notice wording for comprehension.

Risks & Trade-offs

What could go wrong, who it affects, and what I accepted

Engineering

- DigiLocker integration & uptime dependency
- Age-detection accuracy (false positives)
- Building an auditable consent ledger

Business

- Added steps may dip under-18 signups short-term
- Ad revenue from minors goes to zero
- Support cost of a new flow

User & edge cases

- Teen with no available parent / guardian
- Declined consent → limited curated mode
- Deletion requests, shared devices, custody disputes

Trade-offs I accepted

Soft-pending lets a teen watch a curated feed before consent completes — I accept a small ungated window to protect learning continuity, bounded by safe-only content. **DigiLocker over broader verification** — I trade some reach for government-grade trust and true data minimisation.

THE PRINCIPLE

**“Consent should be as easy
to withdraw as it is to give.”**

Verified Family Consent brings YouTube India in line with the DPDP Act by fixing consent at the root — a verified guardian, privacy by default, and full data rights — while keeping the teen learning throughout.

Deliverables: **1 Product Thinking Deck (this)** **2 Clickable Prototype (index.html)**

Thank you · Harsh Tak · Vedantu PM Intern Assignment